

■ 2.1.8 Selecting Parts of Expressions Based on Criteria

Section 1.2.4 showed how you can pick out elements of lists based on their *positions*. Often, however, you will need to select elements based not on *where* they are, but rather on *what* they are.

`Select[list, f]` selects elements of *list* using the function *f* as a criterion. `Select` applies *f* to each element of *list* in turn, and keeps only those for which the result is `True`.

This defines `gt4` to test whether an expression is greater than 4.

```
In[1]:= gt4[e_] := e > 4
```

This selects the elements of the list for which `gt4` yields `True`, i.e. those which are numerically greater than 4.

```
In[2]:= Select[{2, 15, 1, a, 16, 17}, gt4]
Out[2]= {15, 16, 17}
```

You can use `Select` to pick out pieces of any expression, not just elements of a list.

This defines the function `xfree[e]` to give `True` if the symbol `x` does not appear anywhere in *e*.

```
In[3]:= xfree[e_] := FreeQ[e, x]
```

This gives a sum of terms involving *x*, *y* and *z*.

```
In[4]:= t = Expand[(x + y + z)^2]
Out[4]= x^2 + 2 x y + y^2 + 2 x z + 2 y z + z^2
```

You can use `Select` to pick out only those terms in the sum that do involve the symbol `x`.

```
In[5]:= Select[t, xfree]
Out[5]= y^2 + 2 y z + z^2
```

`Select[expr, f]` select the elements in *expr* for which the function *f* gives `True`

Selecting pieces of expressions.

Section 2.3.2 discusses some “predicates” that are often used as criteria in `Select`.